Unit: 6 Unit 8: Quadratics **Name:** ______________

Topic: Converting Forms (Completing the Square) **Date:** __________

Advanced (Habanero)

Q1. Convert $x^2 + 6x + 8$ to vertex form
(A) $(x+3)^2 - 1$
(B) $(x+3)^2 + 1$
(C) $(x+4)^2 - 8$
(D) $(x+4)^2 + 8$
(E) $(x+3)^2 - 9$

Q2. Complete the square for $x^2 - 4x$
(A) $x^2 - 4x + 4$
(B) $x^2 - 4x - 4$
(C) $(x-2)^2$
(D) $(x+2)^2$
(E) $x^2 + 4x$

Q3. What is the vertex of the parabola $x^2 + 2x + 1$?
(A) $(-1, 0)$
(B) $(-1, 1)$
(C) $(-2, 0)$
(D) $(-2, 1)$
(E) $(-1, -1)$

Q4. Convert $x^2 - 8x + 12$ to vertex form
(A) $(x-4)^2 - 4$
(B) $(x-4)^2 + 4$
(C) $(x+4)^2 - 12$
(D) $(x+4)^2 + 12$
(E) $(x-4)^2 - 16$

Q5. What is the equation of the parabola with vertex $(-2, 3)$ and x-intercept $(1, 0)$?
(A) $x^2 + 4x + 3$
(B) $x^2 + 4x - 3$
(C) $x^2 - 4x + 3$
(D) $x^2 - 4x - 3$
(E) $(x+2)^2 - 3$

Q6. Complete the square for $x^2 + 10x$
(A) $x^2 + 10x + 25$
(B) $x^2 + 10x - 25$
(C) $(x+5)^2$
(D) $(x-5)^2$
(E) $x^2 - 10x$

Q7. What is the axis of symmetry of the parabola $x^2 + 6x + 5$?
(A) $x = -2$
(B) $x = -3$
(C) $x = -1$
(D) $x = 1$
(E) $x = 3$

Q8. Convert $x^2 - 2x - 6$ to vertex form
(A) $(x-1)^2 - 7$
(B) $(x-1)^2 + 7$
(C) $(x+1)^2 - 6$
(D) $(x+1)^2 + 6$
(E) $(x-1)^2 - 8$

Open Ended Questions (FRQ)

Q9. Prove that the vertex form of a parabola is $a(x-h)^2 + k$, where (h, k) is the vertex.

Q10. Find the equation of the parabola with vertex $(-3, 2)$, axis of symmetry $x = -3$, and y-intercept $(0, 5)$.

Chef's Tips

- Provide a clear and concise prompt for each question
- Ensure that the questions are evenly spaced and easy to read
- Allow students to use a calculator for calculations, but not for completing the square